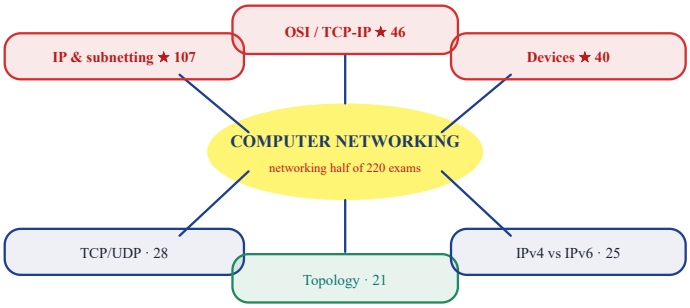
Computer Networking

every networking question — addressing, subnetting, OSI/TCP-IP, devices, TCP/UDP, routing, media — asked in BCS · NTRCA · BPSC · bank · BUET-set papers. (Security & crypto: companion set.)



↑ the number is how many different exams asked that topic (red = most asked)

How to read a question card

> **Q:** The question, as printed in the paper (typos fixed).
> ✎ **Asked in:** the exams it was taken from, e.g. 46th BCS CSE · BPSC AP 2023

Every subnetting answer and delay numerical was **computed and checked**; green = the real result. Several printed subnet answers were cut off or wrong — all recomputed.

Most asked

① IPv4 vs IPv6 · 5
② subnet range of a /28 · 4
③ TCP 3-way handshake · 4
④ TCP vs UDP · 4
⑤ OSI / TCP-IP model & layers

Syllabus coverage

"Data communication, **IP addressing & subnetting**, **OSI & TCP/IP**, network **devices** & switching, **TCP/UDP**, routing, topology, media, IPv4/IPv6, LAN/MAN/WAN, error control." Security & cryptography: companion set.

> **Legend:** **yellow** = key term · **red circle** = must-memorise · **purple box** = memory trick · **red box** = exam trap · **green** = checked answer

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1 IPv4 address: classes, ranges, network/host

Q1: What are IP classes and how do you identify the class of an IP? / Range of class A, B, C. / Identify the class, network and broadcast of 1.2.3.4, 130.1.2.3, 220.15.1.10, 200.1.10.100.

✎ **Asked in:** Parliament Secretariat Programmer 2023 · DESCO AE(IT) 2022 · BPSC AE 2022 · Combined Bank Officer(IT) 2020

Class	1st octet	Default mask	Net/Host bits	Use
A	1–126	255.0.0.0 (/8)	8 / 24	huge networks
B	128–191	255.255.0.0 (/16)	16 / 16	medium
C	192–223	255.255.255.0 (/24)	24 / 8	small (254 hosts)
D	224–239	—	—	multicast
E	240–255	—	—	experimental

127 is reserved for **loopback** (127.0.0.1 = localhost), so class A usable is **1–126** (and 0 is "this network"). **Class C max = 254** usable hosts (BPSC AME 2019). Worked identification: 1.2.3.4 → class A, net 1.0.0.0, directed bcast 1.255.255.255; 130.1.2.3 → class B, net 130.1.0.0, bcast 130.1.255.255; 220.15.1.10 → class C, net 220.15.1.0, bcast 220.15.1.255; 200.1.10.100 → class C, net 200.1.10.0, bcast 200.1.10.255. Limited broadcast is always **255.255.255.255**.

MCQ as asked	answer	asked in
Which class gives a max of only 254 hosts per network?	**Class C**	BPSC AME 2019
Class C IP address is for a ___-bit network	**24-bit** network ID	Combined Bank Officer(IT) 2020
Network address for 178.112.13.10/8	**178.0.0.0**	Combined Bank AHE/AE(IT) 2021
Which is a private IP? 192.169.10.10 / 11.5.10.10 / 1.1.1.1 / **172.16.5.3**	**172.16.5.3**	DoICT ANE 2019

Exam trap: Class A is **1–126**, not 1–127: 127.x.x.x is loopback. 0.x and 255.x are reserved. "Directed broadcast" = all host bits 1 within a network; "limited broadcast" = 255.255.255.255.

Memory trick: First-octet ranges: **A < 128, B 128–191, C 192–223, D 224–239, E 240+**. Class by the leading bits: 0=A, 10=B, 110=C, 1110=D. Default masks: /8, /16, /24.

2 Private vs public, loopback, static vs dynamic

Q2: Give the private IP ranges of class A, B and C (with max hosts). / Difference between public and private IP. / What is NAT and why is it used?

✎ **Asked in:** BBA AME 2025 · NSDA AME 2024 · BREB AP 2023 · Rupali Bank ANA 2023 · Combined Bank AHE 2021

Class	Private range	CIDR	Hosts
A	10.0.0.0 – 10.255.255.255	**10.0.0.0/8**	~16.7 M
B	172.16.0.0 – 172.31.255.255	**172.16.0.0/12**	~1 M
C	192.168.0.0 – 192.168.255.255	**192.168.0.0/16**	65,536

Private IPs are not routable on the internet; **NAT** (Network Address Translation) maps many private IPs to one public IP, saving addresses and hiding the internal network. A network can reach the internet without NAT only if every host has a public IP (e.g. IPv6). **Static IP** = fixed, set by hand (servers); **dynamic** = assigned by **DHCP**, changes. **127.0.0.1 / localhost** = loopback, tests the host's own TCP/IP stack (`ping 127.0.0.1`).

Q3: Two services needed so a computer gets a dynamic IP and reaches the internet by domain name? / Which protocol auto-assigns IP addresses? / Which maps an IP to a MAC address?

✎ **Asked in:** BB AME 2017 · BTCL JAM 2022 · NESCO AM(ICT) 2021 · DoICT ANE 2019

DHCP (gives IP, mask, gateway, DNS server — Combined Bank SO(IT) 2021) + **DNS** (name → IP). **ARP** maps a known IP → MAC (DoICT ANE 2019, Combined Bank AHE/AE(IT) 2021); **RARP** the reverse. NAT lets private hosts share a public IP.

MCQ as asked	answer	asked in
Which is the loopback address? 255.255.255.0 / **127.0.0.1** / 255.0.0.0 / 127.127.127.0	**127.0.0.1**	Combined Bank AHE/AE(IT) 2021
A client-server program giving IP, mask, router & name-server to a host	**DHCP**	Combined Bank SO(IT) 2021
Maps a known IP address to a MAC address	**ARP**	Combined Bank AHE/AE(IT) 2021
What stays local on the LAN and never crosses a router?	**MAC address**	BUET IICT April 2021

3 Subnet mask, CIDR & why we subnet

Q4: What is a subnet and a subnet mask? Why is it used? / What is CIDR? / Primary motivation for moving from classful to classless (CIDR) addressing?

✎ **Asked in:** Combined Bank SO(IT) 2024 · Rupali Bank ANE 2023 · BPDB AE(CSE) 2021 · BPSC AP 2021 · Sonali Bank ADBA 2024 · Combined Bank ADA 2024

A **subnet mask** separates an IP into **network** (1-bits) and **host** (0-bits) parts; it is why a device knows which addresses are "local". **Subnetting** splits one big network into smaller ones — less broadcast traffic, better security/management, no wasted addresses. **CIDR** (Classless Inter-Domain Routing, written `/n`) drops fixed classes so any prefix length is allowed — the motivation was to **stop IPv4 address exhaustion** and shrink routing tables (via supernetting).

CIDR	Mask	Block size	Usable hosts
/24	255.255.255.0	256	254
/25	255.255.255.128	128	126
/26	255.255.255.192	64	62
/27	255.255.255.224	32	30
/28	255.255.255.240	16	14
/30	255.255.255.252	4	2 (point-to-point)

Memory trick: **Block size = 256 − mask octet**. For /26 the interesting octet is 192, block = 256−192 = **64**; subnets start at 0, 64, 128, 192. Usable = block − 2 (network + broadcast). Subnets = 2^(borrowed bits); hosts = 2^(host bits) − 2.



↑ the block size (here 64) is the jump between consecutive subnets

Exam trap: The last mask octet is always one of **128, 192, 224, 240, 248, 252, 254** (the "magic numbers"). /30 gives exactly 2 usable (a router-to-router link); /31 is used point-to-point with no host subtraction.

4 Find network / broadcast / range (worked)

Q5: Given IP 192.168.1.50, mask 255.255.255.240: find the network address, broadcast address and valid host range.

✎ **Asked in:** BIWTA AE(CSE) 2023 · BPDB AE(CSE) 2024 · BRICM AME 2024 · BTCL AME 2023 · NWPGCL AM(ICT) 2024

/28 → block 16; 50 falls in the 48–63 block. Network **192.168.1.48**, hosts **.49–.62**, broadcast **192.168.1.63**, mask 255.255.255.240, 14 usable. (The printed answer was cut off.)

Problem	Network · Broadcast · Range (computed)	asked in
192.168.100.128/26	net **.128**, bcast **.191**, hosts **.129–.190**, 62 usable (printed .129–.191 wrong)	GTCL AE(CSE) 2022
192.162.10.140/26	net **.128**, bcast **.191**, hosts **.129–.190**, mask .192	BSCL AE(CSE) 2026
192.168.111.169/28	net **.160**, bcast **.175**, first **.161**, last **.174**	BBA AME 2025
172.16.128.120/25	mask .128, net **172.16.128.0**, bcast **.127**, 126 usable	46th BCS CSE · APSCL AE(ICT) 2021 · NACTAR 2020
192.168.5.154/27	net **.128**, first **.129**, last **.158**, bcast **.159**, mask .224	NSDA AME 2024
172.16.236.92/27	mask .224, net **172.16.236.64**, bcast **.95**	NWPGCL JAM(ICT) 2022
172.168.19.92/27	mask .224, net **172.168.19.64**, bcast **.95**, 30 hosts	DUET AP 2025
192.168.2.64/26	mask .192, net **.64**, bcast **.127**, first **.65**, last **.126**, 62 usable	BUET ICT Cell AHE 2025
172.16.10.0/25	net **172.16.10.0**, range **.1–.126**, bcast **.127**	Parliament Secretariat SAE 2023
192.168.5.154/26	net **.128**, mask .192 (NSDA AP 2022)	NSDA AP 2022
192.168.10.2/28	mask .240, net **.0**, first **.1**, last **.14**, bcast **.15**	BCC AP 2021
180.79.35.5/24	class B host? → net **180.79.35.0**, bcast **.255**, 254 usable	PGCB SAE(Computer) 2020

Method: (1) block = 256 − mask octet; (2) network = largest multiple of block ≤ the IP's octet; (3) broadcast = network + block − 1; (4) first = net+1, last = bcast−1. Also asked for the same /26 mask: NSDA AP 2022, BPSC AP 2021, Fisheries & Livestock Instructor(CS) 2023, BPSC SAE 2021, Sonali Bank ADA 2026.

5 Larger blocks & host counts (worked)

Problem	Network · Broadcast · usable (computed)	asked in
172.16.128.120/25	mask .128, net 172.16.128.0, bcast .127, **126**	46th BCS CSE
192.168.10.0/23	mask 255.255.254.0, bcast 192.168.11.255, last usable **192.168.11.254**, **510**	BTCL AM(Technical) 2021 · SGFL AGE 2021
172.16.128.123/22	net 172.16.128.0, mask 255.255.252.0, bcast 172.16.131.255, **1022**	MoFood Net/Web Mgr 2025
10.2.1.3/22	mask 255.255.252.0, net 10.2.0.0, bcast 10.2.3.255, **1022**	BPSC AE 2022 · Fisheries Instructor(CS) 2023
105.38.89.234/20	net 105.38.80.0, bcast 105.38.95.255, **4094**	Combined Bank AHE 2021
172.192.20.10/20	network **172.192.16.0**	BUET IICT April 2021
192.168.16.0/20	**4096** addresses (block 16 in 3rd octet)	49th BCS CSE (2025)
168.20.96.63, mask 255.255.192.0 (/18)	net 168.20.64.0, bcast 168.20.127.255, **16382**	Petrobangla AM(IT) 2022 · BAERA Engineer(CSE) 2023
124.67.48.82, mask 255.255.224.0 (/19)	net **124.67.32.0**, bcast 124.67.63.255	BAERA Engineer(CSE) Jan 2021
172.16.0.0/19 (also 172.10.0.0/19)	8 subnets × 8190 hosts (borrow 3 bits from /16)	Rupali Bank ANA/ANE 2023
172.16.99.45 classful (class B /16)	mask 255.255.0.0, net 172.16.0.0, bcast 172.16.255.255	BPSC AP 2023
10.100.108.39, mask 255.255.254.0 (/23)	**512** addresses, first 10.100.108.0, last 10.100.109.255	Parliament Secretariat SAE 2023
10.2.3.20/22	net 10.2.0.0, mask 255.255.252.0, 1022 hosts	BTCL AE(CSE) 2017

/26 usable = 62 (asked many times: NSDA AP 2022, BPSC AP 2021, Fisheries & Livestock Instructor(CS) 2023, BPSC SAE 2021 CIDR block, GTCL AE(CSE) 2022). **Class-default reminder:** a /28 on a class-C network borrows 4 bits ⇒ **16 subnets × 14 hosts** (not "4 subnets" — the 38th BCS (2018) answer sheet printed that wrong). A /20 on class B borrows 4 bits ⇒ 16 subnets (Petrobangla ANE suggestion).

> **Exam trap:** "Usable hosts" always subtracts **2**. For 260 hosts you need ≥ 9 host bits (/23 = 510), so a /24 (254) is too small (PGCB AE(CSE) 2021). Read whether the question wants *total* or *usable*.

Same method, also asked in: ICB Securities AP 2026 · MoL AME 2023 · BGDCL AM(CSE) 2024 · BRPL AM(ICT) 2024 · BOF AP 2022 · Combined Bank AME 2026 · Fisheries & Livestock Instructor(CS) 2023 · MoFood Network/Website Manager 2025 · Combined Bank SO(IT) 2026 · BPSC ME 2021.

6 Divide a network into N subnets (equal & VLSM)

Q6: Divide 192.168.1.0/24 into 4 equal subnets: new mask (CIDR + dotted) and the first usable host of each. / Also 192.168.10.0/24 with borrowed bits = 3.

✎ **Asked in:** Sonali Bank ADA 2026 · Combined Bank AME/AE(IT) 2026(Jan) · BUET AP 2025

4 subnets → borrow 2 bits → **/26, mask 255.255.255.192**, 62 usable each. Subnets: 192.168.1.0/26, .64/26, .128/26, .192/26 (first usable .1, .65, .129, .193). Borrowed = 3 → /27, 8 subnets × 30 hosts.

Problem	Answer (computed)	asked in
10.10.0.0/16 into 8 equal; 3rd subnet first/last	/19, mask 255.255.224.0; 3rd = **10.10.64.0/19**, first .64.1, last .95.254	DPDC AM(ICT) 2025
172.168.0.0/28 into 4	/30; **.0/30, .4/30, .8/30, .12/30**	RAKUB ME/NSE 2021
200.1.2.0/24 into 4 (and into 3)	/26; .0, .64, .128, .192 /26 (use 3 of them)	MRA SCO 2023 · Hydrocarbon AD(ICT) 2025
192.178.10.0/24 into 4	/26; .0/.64/.128/.192 — 254/4 ≈ 62 hosts each	BUET AP 2025
172.168.10.0/24 into 32 subnets	borrow 5 → /29, 8 addr each (6 usable); subnet-1 .0–.7, subnet-32 .248–.255	JBL SO(ANE) 2020
10.2.1.3 into "C-mask /26", usable	mask 255.255.255.192, 62 usable	BPSC SAE 2021 · BPSC AP 2021

VLSM (variable-size, assign biggest block first): e.g. **/22 for 250/250/125/120/60 hosts** (ICB Securities AP 2026): 250→/24 (×2), 125→/25, 120→/25, 60→/26 — allocate largest first so blocks don't overlap. **14.24.74.0/24 → A 120 + B 60** (Combined Bank SO(IT) 2026): A = .0/25 (126), B = .128/26 (62). Choose the smallest block that fits each requirement. Same skill: 192.168.0.0 → A 110, B 50, C 20, D 8 hosts ✎ **Asked in:** Cadet College Lecturer(ICT) 2025

Memory trick: **Subnets need N → borrow ⌈log₂N⌉ bits**; hosts need H → leave ⌈log₂(H+2)⌉ host bits. Equal subnetting: one fixed new mask. VLSM: a different mask per subnet, largest requirement first.

Exam trap: In VLSM, always allocate the **biggest** requirement first, or blocks overlap. "4 equal subnets" borrows **2** bits (not 4). Round each requirement **up** to the next power of two before choosing the mask.

More "divide into N subnets" papers: BUET M.Sc. April 2024 · BUET M.Sc. May 2019 · BUET M.Sc. May 2016 · BUET M.Sc. April 2017 · PGCB AE(CSE) 2021 · NRCC AP 2021 · BAUST AP 2021 · Rupali Bank ANE 2021 · JBL ASA 2021 · BSCPLC AM(Engineering) 2024.

7 Supernet (CIDR aggregation) & longest-prefix match

> **Q7:** An ISP has 192.168.0.0/24, .1.0/24, .2.0/24, .3.0/24. Aggregate them into one route (CIDR). / What subnet mask combines 192.168.5.0, .6.0, .7.0?
> ✎ **Asked in:** Combined Bank ADA 2026 · Combined Bank Officer(IT) 2026 · BUET IICT April 2021

Supernetting = merging contiguous networks into one shorter prefix to shrink routing tables. Four /24s (.0–.3) share the first 22 bits → **192.168.0.0/22**. For .5/.6/.7: they fall in the .4–.7 block → **192.168.4.0/22**, mask **255.255.252.0** (BUET IICT April 2021 — a /30 can't hold three /24s). The /22 covers .4, .5, .6, .7.

> **Q8:** Routing table (all /23, mask 255.255.254.0): 164.0→I0, 166.0→I1, 168.0→I2, 170.0→I3, default→I4 (172.168.x). Find the interface for 165.121, 167.151, 163.151, 171.92.
> ✎ **Asked in:** BAPEX AGM(ICT) 2023

/23 groups the 3rd octet in pairs (164–165, 166–167, 168–169, 170–171): 172.168.165.121 → **I0**, .167.151 → **I1**, .171.92 → **I3**; 172.168.163.151 matches no entry → **default I4**. (The printed "I2" for 163.151 is wrong — 163 is outside every /23 range.) A router uses the **longest-prefix match**: for 192.168.20.45 with routes /16, /20, /22, /25 the most specific that matches wins (49th BCS CSE (2025)).

Problem	Answer	asked in
CIDR of the range 10.16.0.2 – 10.16.3.255	**10.16.0.0/22** (1024 addresses)	BCIC AP 2025
245.248.128.0/20: half to Admin, ¼ to HR	Admin /21 (2048), HR /22 (1024), rest /22	Dhaka WASA AME(Network) 2025
192.168.3.0 needs 254 usable: CIDR & mask	**/24, 255.255.255.0**	BB AD(ICT) 2025
Combine to one route: 3 contiguous /24s → mask	**255.255.252.0 (/22)**	BUET IICT April 2021

How to supernet: write the networks in binary, find the longest common prefix, that length is the supernet mask. The networks must be **contiguous** and the block must start on a valid boundary (e.g. .4/22 not .5/22). **Longest-prefix match**: when several routes match a destination, the router picks the one with the most 1-bits in the mask (most specific), and a **default route** (0.0.0.0/0) catches everything else.

> **Memory trick:** **Subnetting borrows host bits (longer prefix); supernetting gives them back (shorter prefix).** Supernetting shrinks the routing table; the match rule is always "longest prefix wins".

8 ARP, DHCP, NAT, ICMP; MAC vs IP vs port

> **Q9:** Identify the protocol: (i) auto-assign IP, (ii) name→IP from a URL, (iii) retrieve email, (iv) IP→MAC, (v) report errors to the source IP.
> ✎ **Asked in:** NESCO AM(ICT) 2021 · BTCL AM(Technical) 2021

Job	Protocol	Job	Protocol
auto-assign IP	**DHCP**	IP → MAC	**ARP**
name → IP	**DNS**	error reporting	**ICMP**
retrieve email	**POP3/IMAP**	share one public IP	**NAT**

> **Q10:** Compare MAC address, IP address and port number in a TCP/IP network. / Convert 192.168.101.5 to binary.
>
> ✎ **Asked in:** Parliament Secretariat SAE 2023 · Rupali Bank ANE 2023 · Rupali Bank ANA 2023

MAC = 48-bit physical (data-link), burned in, local only; **IP** = 32-bit logical (network), routable, can change; **port** = 16-bit (transport), identifies the application/process. 192.168.101.5 = `11000000.10101000.01100101.00000101`. **NAT** translates private↔public IP; **OSPF** delivers packets by shortest path at the network layer (Combined Bank AP 2026, Bank AP written (bank name cut off on the sheet)).

MCQ as asked	answer	asked in
Valid host IDs for 192.168.2.1/23 LAN? (.1.100 / .1.198 / .2.255 / .3.0)	**.1.198 and .3.0** (block 192.168.2.0/23 = .2.0–.3.255; 192.168.1.x is a different block)	BUET M.Sc. April 2021
Valid hosts for 192.166.2.16/28	**14**	BUET IICT April 2021
Max subnets if each holds ≤14 hosts (from a /24)	**16** (/28)	BUET IICT April 2021
Class B, mask .248 (/21): hosts per subnet	**2046**	Combined Bank SO(IT) 2021
Purpose of NAT	**Map private IPs to a public IP** (save addresses, hide LAN)	Combined Bank Officer(IT) 2026
Layer-3 routers vs switches for inter-department forwarding (small org)	**Layer-3 switch** (faster in-LAN routing)	Combined Bank ADA 2021

> **Memory trick:** **DHCP** gives an address, **DNS** resolves a name, **ARP** finds the MAC, **ICMP** reports errors (ping/traceroute), **NAT** shares a public IP. MAC = L2 physical, IP = L3 logical, port = L4 process.

More IP / addressing papers: BTV AP 2019 · PSB AP 2018 · NESCO AM(ICT) 2018 · BWDB AP 2018 · BCPCL JI 2019 · DUET AP 2020 · BAERA Engineer(CSE) January 2021 · BGDCL AM(CSE) 2024 · Combined Bank AME 2026.

9 OSI model: 7 layers, order, PDUs

> **Q11:** What is the OSI model? List the layers top-to-bottom. / Describe each layer. / Which layers handle data transfer vs user interaction?
>
> ✎ **Asked in:** BTRC SAD 2021 · DESCO AE 2023 · RAKUB NSE 2021 · BARC Programmer 2023 · BPSC AP 2023 · Combined Bank AP(SB & Janata) 2026

#	Layer	Function	PDU	Device/Protocol
7	**Application**	user services	data	HTTP, DNS, SMTP
6	Presentation	encrypt, compress, format	data	SSL/TLS, JPEG
5	Session	start/manage/end sessions	data	NetBIOS, RPC
4	**Transport**	end-to-end, ports, reliability	**segment**	TCP, UDP
3	**Network**	logical addressing, routing	**packet**	IP, router
2	**Data link**	MAC, framing, error detection	**frame**	switch, bridge
1	Physical	bits on the medium	**bit**	hub, cable, NIC

Order (top→bottom): All People Seem To Need Data Processing. **PDU chain:** data → **segment** (L4) → **packet** (L3) → **frame** (L2) → **bit** (L1) (Combined Bank SO(IT) 2023). **Flow control** happens at the **data-link** and **transport** layers (BB AP 2019). The **network layer** is where routing and the IP packet live (BPSC AME 2019).

Each layer's one-line job: Application — network services to apps (browser, mail); **Presentation** — translation, encryption, compression; **Session** — dialogue control (open/keep/close); **Transport** — segmentation, ports, reliable end-to-end delivery; **Network** — logical (IP) addressing + best-path routing; **Data link** — framing + MAC addressing + error detection; **Physical** — bits as signals on the medium. Layers 1–3 are "network/media support", 5–7 are "user support", and layer 4 joins them.

> **Memory trick:** Top→bottom: **A**ll **P**eople **S**eem **T**o **N**eed **D**ata **P**rocessing. Bottom→top: **P**lease **D**o **N**ot **T**hrow **S**ausage **P**izza **A**way. PDUs: bit → frame → packet → segment → data.

> **Exam trap:** A **firewall** works mainly at L3 (packet filter) — can go up to L7; **router** at L3; **switch/bridge** at L2; **hub/repeater** at L1; **access point** at L2. End-to-end encryption best fits the presentation layer (L6).

10 OSI layer functions, protocols & devices (MCQ-heavy)

> **Q12:** State the layer at which each works: CSMA, OSPF, TCP, RIP, UDP. / Name the device at each OSI layer. / Which two layers are "flow control"?
> ✎ **Asked in:** BPDB AE(CSE) 2023 · 18th NTRCA School-2 (2024) · BB AP 2019

CSMA → data link; **OSPF, RIP, IP, ICMP** → network; **TCP, UDP** → transport (BPDB AE(CSE) 2023). Devices by layer: L1 hub/repeater/NIC, L2 switch/bridge/AP, L3 router/L3-switch, L7 gateway (18th NTRCA School-2 (2024)). Session layer establishes/manages/terminates logical connections (49th BCS CSE (2025)); data-link's primary job is error detection/correction + framing (49th BCS CSE (2025)).

MCQ as asked	answer	asked in
Layer responsible for port binding (TCP/IP)	**Transport**	49th BCS CSE (2025)
Layer that establishes/manages/terminates logical connections	**Session**	49th BCS CSE (2025)
Data-link layer's primary function	**Error detection & correction** (+ framing)	49th BCS CSE (2025)
A firewall works at which OSI layer?	**Network** (L3 packet filter; up to L7)	49th BCS ICT (2025)
Which layer delivers packets to the port, sender→destination?	**Transport**	BCC AP 2021
OSI model has how many layers?	**7**	Combined Bank Officer(IT) 2020
Bridges are used at which layer?	**MAC / data-link (L2)**	BUET IICT April 2021
Main function of the transport layer	**Transparent end-to-end transfer between end users**	Combined Bank SO(IT) 2021
An access point operates at which layer?	**Data link**	Combined Bank SO(IT) 2021
Pairs of transport-layer protocols: IP,ICMP / ARP,IP / TCP,IP / UDP,IP	**None correct** (IP is network; intended answer TCP,UDP not offered)	BPSC AME 2022

Protocol-to-layer cheat sheet: L7 HTTP/FTP/DNS/SMTP/POP3/DHCP/SNMP · L4 TCP/UDP · L3 IP/ICMP/IGMP/OSPF/RIP · L2 Ethernet/PPP/ARP/switch · L1 cables/hub. (ARP straddles L2/L3.) **Why layering?** modularity — each layer can change independently, and encapsulation hides lower-layer detail from upper layers.

> **Memory trick:** Memorise the **device-per-layer** ladder (hub L1 → switch L2 → router L3) and the **PDU** ladder (bit → frame → packet → segment); most OSI MCQs are one of these two.

11 TCP/IP model: 4 layers, protocols, devices

> **Q13:** Explain the TCP/IP model with its layers, protocols and devices. / Tabular form: each TCP/IP layer, its function, protocols, device and software.
>
> ✎ **Asked in:** BRICM AME 2024 · RAKUB NSE 2021 · SPARRSO 2020 · 40th BCS (2020) · EGCB SAE(ICT) 2023 · Combined Bank AME/AE(IT) 2024 · Hydrocarbon Unit AD(ICT) 2025

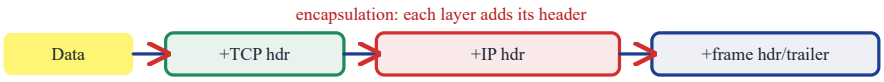
TCP/IP layer	OSI equivalent	Function	Protocols / device
Application	App+Pres+Session (5–7)	user services	HTTP, FTP, DNS, SMTP
Transport	Transport (4)	end-to-end, ports	**TCP, UDP**
Internet	Network (3)	addressing, routing	IP, ICMP, ARP · router
Network access	Data link + Physical (1–2)	framing, bits on wire	Ethernet · switch, NIC

The TCP/IP (DoD) model has **4 layers**; it maps to OSI's 7 by merging the top 3 into Application and the bottom 2 into Network-access. **MAC-layer functions:** framing, physical (MAC) addressing, media access control, error

detection (Rupali Bank ANA 2023). **Two transport protocols:** TCP (reliable — HTTP, FTP, email) and UDP (fast — DNS, streaming, VoIP) (Parliament Secretariat SAE 2023). **TCP/IP tunnelling** wraps one protocol's packet inside another (VPN/IPv6-over-IPv4) (BPSC AP 2023).

> **Q14:** What is a protocol? / Which layer links the network-support and user-support layers? / Data units of Physical, Data-link, Network, Transport layers.
> ✎ **Asked in:** SPARRSO 2020 · BCC AP 2023 · Jamuna Oil JO(IT & MIS) 2025

A **protocol** = agreed rules for communication. The **transport layer (4)** links the lower network-support layers (1–3) to the upper user-support layers (5–7). Data units: Physical = **bit**, Data-link = **frame**, Network = **packet**, Transport = **segment**.



↑ data → segment → packet → frame → bits; the receiver strips each header (de-encapsulation)

> **Memory trick:** TCP/IP = **A-T-I-N** (Application, Transport, Internet, Network-access). Only two models get tested: OSI (7, theoretical) and TCP/IP (4, actually used on the internet).

12 OSI vs TCP/IP; ARP's role; protocol table

> **Q15:** Difference between the OSI and TCP/IP models. / Compare and contrast. / Difference between DoD and OSI.
> ✎ **Asked in:** Combined Bank SO(IT) 2023 · RAKUB ANSE 2023 · Rupali Bank ANA 2023 · SPCBL SAME 2022 · DESCO AE(IT) 2022 · SPCBL AME 2021

Point	OSI	TCP/IP
layers	**7**	**4**
basis	theoretical reference	practical, built the internet
model vs protocol	protocol-independent	protocol-specific
approach	vertical / strict	horizontal / flexible
session & presentation	separate layers	merged into Application

> **Q16:** Discuss the main role of ARP in the network layer. / Fill the protocol table (protocol + full form) for: send mail to a mail server; auto-assign IP; IP→MAC; report errors.
> ✎ **Asked in:** BPSC AP 2023 · BTCL AM(Technical) 2021

Description	Protocol (full form)
send mail from sender to the receiver's mail server	**SMTP** (Simple Mail Transfer Protocol)
dynamically assign IP addresses	**DHCP** (Dynamic Host Configuration Protocol)
resolve an IP address to a MAC address	**ARP** (Address Resolution Protocol)
report errors to the source IP of a packet	**ICMP** (Internet Control Message Protocol)
retrieve email from a mail server	**POP3 / IMAP**

ARP lets a host find the MAC address for a known IP on the same LAN (broadcast "who has this IP?", unicast reply), so the frame can be addressed at L2 — it bridges the network and data-link layers.

> **Exam trap:** OSI is the **reference** everyone studies; TCP/IP is what actually runs. The internet uses TCP/IP, not OSI. ARP is often listed at the network/data-link boundary — don't call it transport.

> **Memory trick:** OSI = **7** layers (model), TCP/IP = **4** (built the net). Map them: OSI 5-6-7 → TCP/IP Application; OSI 1-2 → TCP/IP Network-access; 3→Internet, 4→Transport stay the same.

OSI / TCP-IP also asked in: 17th NTRCA CSE (2023) · 16th NTRCA ICT · BWDB AP 2018 · BTRC DAD 2021 · NWPGCL JAM 2020 · WZPDCL AE(CSE) 2019 · NWPGCL AE(CSE).

13 Hub, switch, router, bridge, gateway, repeater

> **Q17:** Describe the devices: Repeater, Hub, Bridge, Switch, Router (and Gateway). / Difference among Hub, Switch and Router. / Difference between Router and Bridge/Modem/Gateway.
> ✎ **Asked in:** BCSCL NE 2017 · Combined Bank SO(IT) 2024 · Combined Bank Officer(IT) 2023 · PGCL SAE 2021 · DESCO SAE 2023 · JBL ANE 2022

Device	OSI layer	What it does	Domains
Repeater	1	regenerates the signal	one collision+broadcast
Hub	1	repeats to **all** ports (dumb)	one collision domain
Bridge	2	connects 2 LAN segments by MAC	splits collision
Switch	2	forwards frames by **MAC**, per-port	**per-port collision**
Router	3	forwards packets by **IP**, joins networks	splits broadcast
Gateway	up to 7	translates between different protocols	—

Switch vs router: a switch moves frames *inside* a LAN by MAC; a router moves packets *between* networks by IP and is where you'd **block a site** (46th BCS CSE). **Gateway vs router:** a gateway can translate protocols, a router only routes IP — a gateway is more flexible across different network types (17th NTRCA CSE (2023), 17th NTRCA ICT (2023)). A **bridge/switch** connects multiple LANs (Combined Bank AHE/AE(IT) 2021). L2 vs L3 switch: L3 can route between VLANs.

MCQ as asked	answer	asked in
Operates at the network layer, connects different networks	**Router**	49th BCS CSE (2025)
Bridge is a ___ device	**Layer-2**	BPSC AME 2019
Queuing delay of the 10th packet (1200 B each, 1 Gbps)	9 × (1200×8/10⁹) = **86.4 µs**	49th BCS CSE (2025)
Device to connect several LANs	**Router / switch / bridge**	Combined Bank AHE/AE(IT) 2021
Short notes: Node, Hub, Backbone, Router, Gateway	**end device · L1 repeater-to-all · main high-speed link · L3 inter-network · protocol translator**	BB AME 2019

Modem (modulator-demodulator) converts digital ↔ analog for a phone/cable line — different from a router, which forwards IP packets (SPARRSO 2020). A **NIC** (network interface card) gives a host its MAC address and physical link. Collision domain = set of devices that can collide (one per hub, one per switch *port*); broadcast domain = reach of a broadcast (one per router interface / VLAN).

Devices / router-gateway also asked in: 17th NTRCA CSE (2023) · 16th NTRCA ICT · BEPZA Programmer 2023 · WZPDCL AE(CSE) 2019 · BPDB AE(CSE) 2018 · BUET M.Sc. May 2016 · DESCO AE(CSE) 2016.

14 Circuit vs packet switching; traceroute; duplex

> **Q18:** Discuss circuit switching and packet switching with diagrams. Which is better and why? How does packet switching work step by step?
>
> ✎ **Asked in:** 45th BCS CSE (2024) · RAKUB ANSE 2023 · Rupali Bank ANA 2023 · Hydrocarbon Unit AD(ICT) 2025 · NPCBL Engineer(Cyber Security) 2026 · BUET AP(Exam Controller) 2025

	Circuit switching	Packet switching
path	dedicated circuit, whole call	**per-packet**, shared links
setup	needed first (telephone)	none; store-and-forward
efficiency	reserves bandwidth (wasteful)	**statistically shares, efficient**
use	PSTN voice	the internet (IP)

Packet switching is better for data: links are shared, no idle reservation, resilient to link failure (packets reroute). Steps: data → split into packets (header + payload) → each routed independently → reassembled at the destination. **Half-duplex** = one direction at a time (walkie-talkie / hub); **full-duplex** = both at once (switch) (Combined Bank ADA 2026).

> **Q19:** A packet crosses 4 routers, each 2 ms processing, 5 ms transmission per link (ignore propagation). Total end-to-end delay? / Traceroute shows 8 hops then "* * *" — how many routers?
>
> ✎ **Asked in:** Combined Bank AME/AE(IT) 2026(Jan) · Titas Gas AE(CSE) 2024

5 links × 5 ms transmission + 4 routers × 2 ms processing = 25 + 8 = **33 ms**. **traceroute** lists each router (hop) on the path; * * * = a hop that doesn't reply to the probe (filtered), not necessarily down.

Message / packet / virtual-circuit switching: message switching stores the whole message at each node (store-and-forward, slow); datagram packet switching routes each packet independently (the internet); virtual-circuit switching sets a logical path first (frame relay, ATM, MPLS) — a middle ground between circuit and datagram. **Connection types** (SPCBL SAME 2022): router ↔ router (serial/Ethernet), router ↔ switch and switch ↔ PC (straight-through), PC↔PC and switch↔switch (crossover), PC↔switch console (rollover).

> **Memory trick:** **End-to-end delay = Σ(transmission + propagation + processing + queuing)** over every hop. Circuit switching reserves bandwidth up front; packet switching shares it and is more efficient for bursty data.

> **Exam trap:** Circuit switching gives a fixed, low delay but holds the circuit even when idle; packet switching shares links but adds a variable queuing delay. For "which is better", name the traffic: voice suits circuits, bursty data suits packets.

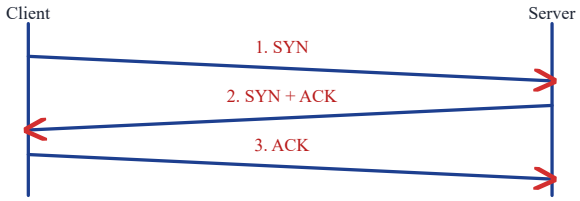
15 TCP vs UDP; 3-way handshake; TCP functions

> **Q20:** Differentiate between TCP and UDP (with examples). / What are the functions TCP performs to make an unreliable network reliable?
>
> ✎ **Asked in:** 37th BCS · 45th BCS CSE (2024) · BPSC AP 2021 · Combined Bank SO(IT) 2024 · Combined Bank AP(SB & Janata) 2026 · Combined Bank Officer(IT) 2020

	TCP	UDP
connection	**connection-oriented** (handshake)	**connectionless**
reliability	ACKs, retransmit, ordered	best-effort, no ACK
header	20 bytes	8 bytes
speed	slower, heavy	**fast, light**
use	HTTP, FTP, SMTP	DNS, DHCP, VoIP, streaming

TCP's reliability functions: connection setup/teardown, sequence numbers (ordering), ACKs + retransmission (lost packets), checksum (errors), flow control (window), congestion control, duplicate removal. DNS mostly uses **UDP** (one quick query/reply; low overhead) and switches to TCP for large/zone transfers (Combined Bank Officer(IT) 2025, BUET M.Sc. April 2021).

> **Q21:** Draw the TCP 3-way handshake for connection setup.
>
> ✎ **Asked in:** BCIC AP 2022 · BGDCL AE(CSE) 2021 · BRICM AME 2024 · Sonali Bank Officer(IT) 2021



↑ SYN → SYN-ACK → ACK: connection established, then data flows

Exam trap: UDP is connectionless but **not** "unreliable by design to be bad" — it trades reliability for speed. FTP uses TCP (DoICT AP 2020). Some services (DNS) use **both** TCP and UDP (Combined Bank SO(IT) 2021).

Memory trick: **TCP = phone call** (set up, talk reliably, hang up); **UDP = postcard** (just send). Handshake SYN→SYN-ACK→ACK; teardown is a 4-way FIN/ACK. Well-known ports: HTTP 80, HTTPS 443, DNS 53, SMTP 25.

TCP/UDP & transport also asked in: DESCO SAE 2022 · BTRC AD 2021 · NHA AP 2019 · 16th NTRCA ICT.

16 Flow vs congestion control; AIMD; ARQ

Q22: Difference between flow control and congestion control. / How does TCP congestion control (AIMD) work? / How does stop-and-wait ARQ give reliable transfer?

✎ **Asked in:** Combined Bank Officer(IT) 2024 · 18th NTRCA College ICT (2024) · JBL ANA 2022 · BPSC AP 2021 · Combined Bank SO(IT) 2022

	Flow control	Congestion control
protects	a slow **receiver**	the **network** core
mechanism	receiver window (rwnd)	congestion window (cwnd), AIMD
scope	end-to-end (2 hosts)	all senders sharing links

AIMD = Additive Increase (cwnd + 1 per RTT while OK), Multiplicative Decrease (halve cwnd on loss) — probes for bandwidth, backs off on congestion. **Stop-and-wait ARQ:** send one frame, wait for its ACK before the next; a lost frame/ACK times out and is resent (sequence numbers avoid duplicates). **Framing, flow control, error control** are data-link functions; reliability is added again at transport (JBL ANE 2022).

Q23: Why does DNS use UDP? Describe the DNS resolution of www.companybd.com. / Which service uses both TCP and UDP?

✎ **Asked in:** Combined Bank Officer(IT) 2025 · Combined Bank SO(IT) 2021

DNS over UDP: one small query + one reply, no handshake overhead → fast; TCP only for big responses/zone transfers. **Resolution:** browser → OS resolver → recursive resolver → **root** → **TLD (.com)** → **authoritative** server → IP returned and cached. **DNS uses both TCP and UDP** (Combined Bank SO(IT) 2021).

TCP congestion phases: slow start (cwnd doubles each RTT until a threshold), **congestion avoidance** (linear +1), on loss → **fast retransmit/recovery** (halve cwnd) or timeout (back to slow start). **Sliding window** lets several unacked frames be in flight = better link use than stop-and-wait; window efficiency = W·Tt / (Tt + 2·Tp). **ARQ types:** stop-and-wait (1 frame), go-back-N (resend from the lost one), selective-repeat (resend only the lost one — most efficient).

Congestion signs & control: rising delay, dropped packets, timeouts. Approaches: **open-loop** (prevent — admission control, traffic shaping with leaky/token bucket) and **closed-loop** (react — AIMD, choke packets, ECN). A stable, low end-to-end latency means the window stays near the bandwidth-delay product with few losses.

> **Memory trick:** **Flow control = don't drown the receiver; congestion control = don't drown the network.** TCP does both; UDP does neither. AIMD = slowly climb, sharply back off.

17 Static/dynamic, distance-vector vs link-state

> **Q24:** What is routing? Static vs dynamic routing (advantages). / Distance-vector vs link-state. / What is an autonomous system?
>
> ✎ **Asked in:** Combined Bank Officer(IT) 2023 · BPSC AE 2022 · RAKUB NSE 2021 · Sonali Bank Officer(IT) 2021 · JBL ANA 2022

	Distance-vector	Link-state
knows	distance+direction to each net	full topology map
shares	whole table with **neighbours**	link states with **everyone**
algorithm	Bellman-Ford	Dijkstra (SPF)
convergence	slow, count-to-infinity	**fast**
example	RIP, IGRP	OSPF, IS-IS

Static routing = routes set by hand: secure, no overhead, but no auto-failover and poor for big networks (WZPDCL SAE 2023). **Dynamic** adapts automatically but uses bandwidth/CPU. An **autonomous system (AS)** = networks+routers under one administration; **BGP** routes between ASes (inter-domain), **RIP/OSPF** within one (intra-domain). **Count-to-infinity** is a distance-vector problem (BPSC AP 2022). **Routing shortest path** uses **Dijkstra** (Combined Bank Officer(IT) 2023).

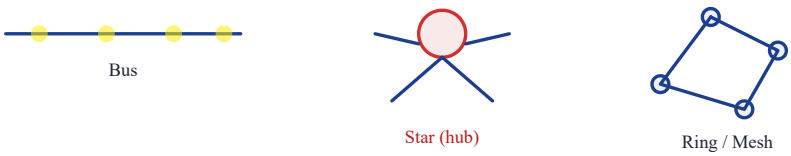
MCQ as asked	answer	asked in
Intra-domain routing protocols pair	**OSPF, RIP**	BPSC AME 2022
Which routing does NOT consume bandwidth?	**Static routing**	BUET IICT April 2021
A routed protocol (carries user data)	**IP** (RIP/OSPF/IGRP are routing protocols)	BUET IICT April 2021
Routing is the major issue for which layer?	**Network layer**	BPSC AME 2019
Full forms: NAT, DSSS, RIP, EIGRP, DHCP, IoT	**Network Address Translation; Direct-Sequence Spread Spectrum; Routing Information Protocol; Enhanced Interior Gateway Routing Protocol; Dynamic Host Configuration Protocol; Internet of Things**	DESCO SAE(Computer) 2026

Loopback / static-routing also asked in: BTCL JAM 2024 (ping 127.0.0.1) · WZPDCL SAE 2023 (static routing + 2 disadvantages). **Static routing drawbacks:** no auto-failover, admin effort grows with the network.

18 Topology types: bus, star, ring, mesh, tree

Q25: What is network topology? Distinguish between Bus, Ring, Tree and Star. Which topology for a university computer lab?

✎ **Asked in:** Rupali Bank ANA 2023 · Rupali Bank ANE 2023 · Combined Bank SO(IT) 2023 · BPSC ADA 2022 · 45th BCS CSE (2024) · Parliament Secretariat SAE 2023



↑ Bus: one backbone · Star: all to a central hub/switch · Ring: loop · Mesh: every-to-every

Topology	Layout	Pros / Cons
Bus	one backbone cable	cheap; one break kills all, collisions
Star	all to a central hub/switch	easy to manage; hub is single point of failure — **used in labs**
Ring	closed loop	orderly; one break affects all (unless dual)
Mesh	every node to every node	very reliable; expensive (lots of cable)
Tree	hierarchy of stars	scalable; root failure is critical

A **university lab** uses **star** (every PC to a central switch) — easy to add/remove a node and a single cable fault isolates one PC. **IEEE 802** is the LAN/MAN standards family (802.3 Ethernet, 802.11 Wi-Fi).

How bus works: all devices share one backbone; a sender broadcasts, every node sees the frame but only the addressee keeps it; terminators at both ends stop signal reflection. Only one device can transmit at a time

(CSMA/CD), so it does not scale. **Logical vs physical topology:** physical = how cables run; logical = how data actually flows (e.g. physical star but logical bus in old Ethernet hubs).

Memory trick: Pick a topology by trade-off: **bus** cheapest, **star** easiest to manage (lab default), **ring** orderly, **mesh** most reliable/costly, **tree** scalable hierarchy.

Exam trap: In a ring, one failed node or link stops the whole ring (a dual ring such as FDDI avoids it); in a star, the central hub or switch is the single point of failure.

19 Links, cables & ports for n devices

Q26: For n devices, how many connections/cable links are needed in Bus, Mesh and Star topology? / 5 devices in a mesh: cables and ports each?

✎ **Asked in:** 15th NTRCA ICT · 17th NTRCA ICT (2023) · BUET M.Sc. April 2021

Topology	Cable links	Ports per device	n = 5
Mesh	**n(n−1)/2**	n − 1	10 cables, 4 ports
Star	n	1 (to hub)	5 cables
Bus	1 backbone (n taps)	1	1 cable
Ring	n	2	5 cables

For 5 mesh nodes: cables = 5×4/2 = **10**, ports = 5−1 = **4** each (BUET M.Sc. April 2021 answer: 10 cables, 4 ports). The **full-mesh** formula n(n−1)/2 is the single most-asked topology numerical. Bus vs ring: bus is cheaper but a backbone break downs everything; ring passes a token so no collisions (16th NTRCA CSE (2019)).

Q27: Name/draw the different topologies with features. / Difference between Star and Ring topology.

✎ **Asked in:** BB IT 2020 · BBL AM(ICT) 2019 · NU AP 2020 · BOF SAE(Computer) 2021 · 15th NTRCA CSE · 13th NTRCA CSE (2016)

Star vs Ring: star is centralised (hub/switch), one node's fault stays local, easy to extend; ring is decentralised (data circulates one way), a single break can stop the loop, but no central bottleneck. Hybrid = tree/mesh+star combos used in real campuses.

n devices	Mesh cables n(n−1)/2	Star/Ring	Bus
4	6	4	1
5	10	5	1
6	15	6	1
n	n(n−1)/2	n	1

Memory trick: **Mesh links = n(n−1)/2** ("handshakes"); each node needs n−1 ports. Star/Ring = n cables. Bus = 1 cable. Mesh is the most reliable and the most expensive.

Topology also asked in: 17th NTRCA CSE (2023) · 16th NTRCA CSE (2019) · BCPCL JI 2019 · BUET M.Sc. May 2019.

20 UTP/STP/coax/fibre; Ethernet standards

Q28: Types of transmission media with advantages/disadvantages. / Compare twisted pair, coax and fibre. / Guided vs unguided; STP vs UTP; why UTP is twisted.

✎ **Asked in:** Combined Bank SO(IT) 2024 · Rupali Bank ANE 2023 · Combined Bank Officer(IT) 2023 · 17th NTRCA ICT (2023) · BPDB AE(CSE) 2021

Media	Type	Speed / distance	Notes
UTP/STP	guided, copper	up to 1–10 Gbps, ~100 m	cheap; EMI (STP shielded)
Coaxial	guided, copper	medium	cable TV, legacy LAN
Fibre optic	guided, glass	**highest, km range**	light, immune to EMI, costly
Wireless	unguided	varies	radio, microwave, satellite

UTP is twisted to cancel electromagnetic interference / crosstalk between the pairs (BPDB AE(CSE) 2021). **Fibre** carries light by **total internal reflection** in the core; highest bandwidth, no EMI, long distance (BPSC AP 2022, MGMCL AM(ICT) 2022). **Guided** = a physical path (cable); **unguided** = air. **Straight-through** cable connects unlike devices (PC↔switch), **crossover** connects like devices (PC↔PC, switch↔switch) (MoL AME 2023). Too-long cables cause **attenuation** and signal loss (BOF AP 2022).

Short Q as asked	answer	asked in
10Base5 / 10Base2 / 10BaseT	**10 Mbps**: thick coax 500 m / thin coax 185 m / twisted pair 100 m	BPSC AP 2021 · RAKUB NSE 2021
Highest-bandwidth media	**Fibre optic**	RAKUB NSE 2021
EPON vs GPON	both passive optical; GPON higher rate (2.5G), EPON Ethernet-based (1G)	15th NTRCA CSE

Fibre parts: core (carries light) → cladding (lower index, reflects light back) → buffer/jacket. **Single-mode** (tiny core, laser, long-haul) vs **multi-mode** (wider core, LED, short/campus). Cat5e ≈ 1 Gbps, Cat6 ≈ 10 Gbps (shorter runs) — a common "Cat5 vs Cat6" distinction (Combined Bank Officer(IT) 2023). **Attenuation** is why cables have a length limit (UTP ~100 m); beyond it, add a repeater/switch.

10BaseT naming: 10 = 10 Mbps, Base = baseband, T = twisted pair (5 = thick coax 500 m, 2 = thin coax ~185 m). Modern: 100BaseTX (Fast Ethernet), 1000BaseT (Gigabit). **Guided** media (copper/fibre) confine the signal to a path; **unguided** (radio, microwave, infrared, satellite) send it through air/space.

Memory trick: **Straight-through for unlike devices, crossover for like devices.** Fibre wins on bandwidth, distance and EMI-immunity; copper (UTP) wins on cost and ease.

Media also asked in: 17th NTRCA CSE (2023) · MoL AME 2023 (straight/cross cable) · MGMCL AM(ICT) 2022 (fibre principle) · BOF AP 2022 (over-length cable).

21 Transmission vs propagation delay (worked)

Q29: Bandwidth vs throughput; latency, throughput, availability. / Transmission time = bits ÷ bandwidth; propagation time = distance ÷ speed.

✎ **Asked in:** 15th NTRCA ICT · 17th NTRCA ICT (2023) · 45th BCS ICT (2024) · 43rd BCS (2023)

Bandwidth = max rate a link *can* carry (bps); **throughput** = what you actually get. **Transmission (Tt)** = packet bits ÷ bandwidth; **propagation (Tp)** = distance ÷ speed of signal.

Problem	Worked answer	asked in
2.5 KB msg, 1 Gbps, 12,000 km, c = 2.4×10^8	Tt = $20000/10^9$ = **0.02 ms**; Tp = $12\times10^6/2.4\times10^8$ = **50 ms**	ICB AMCL AP 2024 · Combined Bank SO(IT) 2021
3 KB email, 1 Gbps, 300 km, c=3×10^8, queue 5 ms	Tt = $24000/10^9$ = 0.024 ms; Tp = 1 ms; one-way ≈ **6.024 ms** (RTT not needed)	BB AD(ICT) 2025
10 Mbps, avg packet 1500 B: max packets/s	$10\times10^6 \div (1500\times8)$ ≈ **833 packets/s** (no answer printed)	BSCL AE(CSE) 2025
Voice 128 kbps, payload 1495 B (1500−5 hdr)	11,960 bits ÷ 128,000 ≈ **93.4 ms** packetization delay	BB AME 2023
TV: 480×500 px, 32 levels (5 bits), 30 pics/s	480×500×5×30 = **36 Mbps**	BB AP 2023
Noiseless 3 kHz, 2 signal levels: max bit rate (Nyquist)	$2\times3000\times\log_2 2$ = **6000 bps**	MM AP 2017
Satellite Tp 250 ms, 1 Mbps, frame 1000 B: stop-wait efficiency, window for 100%	Tt=8 ms; $\eta = 8/(8+500) \approx$ **1.57%**; min window = 1+2Tp/Tt ≈ **64**	RAKUB ANSE 2026

Delay types on one hop: transmission (push bits onto the link = bits ÷ rate), **propagation** (travel the distance = dist ÷ speed), **processing** (router examines the header), **queuing** (wait in the buffer). **Bandwidth-delay product** = bandwidth × RTT = how many bits can be "in flight" (sizes the window). **Utilisation** of stop-and-wait = Tt ÷ (Tt + 2·Tp); sliding window raises it to W·Tt ÷ (Tt + 2·Tp).

Exam trap: "Propagation time" asks distance ÷ speed; the quick bits ÷ bandwidth answer is the **transmission** time (the printed key swapped them in Combined Bank SO(IT) 2021 — 50 ms, not 0.02 ms). Watch the units (KB vs Kb, km vs m).

22 LAN/MAN/WAN; client-server vs P2P; Internet

Q30: Define networking and internetworking. Types of network (LAN/MAN/WAN). / LAN vs WAN. / Client-server vs peer-to-peer. / Intranet vs extranet.

✎ **Asked in:** BPSC ADA 2022 · BPSC ANE 2020 · SPARRSO 2020 · BPSC Lecturer 2017 · BB AME 2016

Type	Scope	Example
PAN	around one person (~10 m)	Bluetooth, hotspot
LAN	one building/campus	office Ethernet/Wi-Fi
MAN	one city	two city branches (BCC AP 2021)
WAN	country / global	the Internet

Internetworking = joining many networks into one (the Internet) via routers and IP (BPDB AE(CSE) 2021). **Client-server**: central servers provide services (scalable, managed); **peer-to-peer**: every node is both client and server (cheap, no central control). **Intranet** = private internal network (company-only); **extranet** = extends the intranet to trusted outsiders (partners). **WWW ≠ Internet**: the Internet is the network; the WWW is one service on it (Teletalk Executive 2019).

MCQ as asked	answer	asked in
Computer-to-computer data transfer technology is called —	**Internet**	13th/14th/15th/18th NTRCA Prelim
Two city branches connected: which network?	**MAN**	BCC AP 2021
Internet over telephone wires (DSL): technology?	**DSL**	BB AD(General) 2021
Workstation can't reach the Ethernet LAN — check first?	**Verify IP configuration**	Combined Bank AHE/AE(IT) 2021
Year the Internet started in Bangladesh	**1996**	17th NTRCA School Prelim (2022)

Internet building blocks: an **ISP** gives you access; an **IX** (Internet Exchange, BD: BDIX) lets local ISPs swap traffic cheaply without going abroad (Combined Bank SO(IT) 2020); an **edge server/CDN** caches content near users for low latency (Petrobangla AM(CSE) 2024). **IGMP** manages multicast group membership (BUET IICT April 2021); **SIP/H.323** set up VoIP calls (Combined Bank SO(IT) 2021).

Memory trick: Scope order: **PAN < LAN < MAN < WAN**. Internet = the network; WWW = one service on it. Intranet = internal only; extranet = internal + trusted partners.

Network-types also asked in: 13th NTRCA School-2 Prelim (2016) · Petrobangla AM(CSE) 2024 (edge server) · Intl Crimes Tribunal Hardware Technician 2024 · BTRC DAD 2021 · Combined Bank SO(IT) 2020 · Teletalk Executive 2019.

23 IPv4 vs IPv6 — the single most-asked NET question

Q31: Differentiate between IPv4 and IPv6. How many bits each? Why is NAT not needed in IPv6?

✎ **Asked in:** BMA AE 2021 · BPSC AME 2021 · BREB AGM(IT) 2021 · RAKUB ANSE 2023 · WZPGCL AE(CSE) 2023 · RPGCL AM(ICT) 2022 · PGCB AE(CSE) 2024

Point	IPv4	IPv6
size	**32 bits**	**128 bits**
addresses	~4.3 billion	~3.4×10^{38}
notation	dotted decimal 192.168.1.1	hex, colons 2001:db8::1
header	20–60 bytes, variable	40 bytes, fixed (faster)
NAT	needed (address shortage)	**not needed** (every host public)
security	optional	IPsec built-in
config	manual / DHCP	auto (SLAAC) + DHCPv6

NAT is unnecessary in IPv6 because the address space is so large every device can have its own public address — no sharing needed. IPv6's **fixed 40-byte header** (no checksum, no fragmentation by routers) means faster processing (RAKUB ANSE 2026). **Example IPv6:** 3FFE:085B:1F1F:0000:0000:0000:00A9:1234, short form 3FFE:85B:1F1F::A9:1234 (38th BCS (2018) — the "::" replaces consecutive zero groups once).

MCQ as asked	answer	asked in
How long is an IPv6 address?	**128 bits**	BPSC AME 2022 · 49th BCS ICT (2025) · DoICT AP 2020
IPv6 basic header is fixed at	**40 bytes**	Combined Bank SO(IT) 2021

Why IPv6 at all: IPv4's ~4.3 billion addresses ran out (world has far more devices). IPv6 also brings built-in IPsec, simpler routing (no fragmentation at routers, no header checksum), autoconfiguration (SLAAC) and better multicast/anycast. **IPv6 vs IPv4 fields:** IPv6 drops the header checksum and the "options" complexity, and renames fields (Traffic Class, Flow Label, Hop Limit instead of TOS/TTL).

Memory trick: **32 → 128** bits is the whole story: 4× the bits = astronomically more addresses, so no NAT, and a fixed header for speed. Written as 8 groups of 4 hex digits, "::" collapses one zero run.

IPv4-vs-IPv6 also asked in: WZPDCL SAE 2023 · BCPCL JI 2019 · WZPDCL AE(CSE) 2019.

24 IPv6 address types, autoconfig, classification

Q32: Classify IPv4 and IPv6 addresses. / Difference between link-local and global unicast (with ranges). / Describe IPv6 stateless autoconfiguration (SLAAC).

✎ **Asked in:** 38th BCS (2018) · Parliament Secretariat SAE 2023 · BSCL AE(CSE) 2026 · 46th BCS ICT · 47th BCS CSE

IPv6 type	Prefix	Meaning
Global unicast	2000::/3	public, internet-routable
Link-local	FE80::/10	on one link only, auto
Unique local	FC00::/7	private (like 192.168)
Multicast	FF00::/8	one-to-group
Loopback	::1/128	localhost

IPv6 has **no broadcast** — it uses unicast, **anycast** (nearest of a group) and multicast. **SLAAC** (stateless autoconfig): host sends a **Router Solicitation (RS)** → router replies with a **Router Advertisement (RA)** carrying the prefix → host builds its own address (prefix + interface ID) → runs **DAD** (Duplicate Address Detection) before using it (BSCL AE(CSE) 2026). IPv4 is classified A–E; IPv6 into unicast/anycast/multicast (38th BCS (2018)).

Q33: What is VLAN? Types (static, dynamic). / What is a VPN?

✎ **Asked in:** RAKUB ANSE 2023 · BPSC AME 2019

VLAN = a logical LAN on a switch that groups ports regardless of location, splitting one physical switch into isolated broadcast domains (static = by port, dynamic = by MAC/policy). **VPN** = a private, encrypted tunnel over the public internet (BPSC AME 2019: "a private network based on a public network"). **Link-local (FE80::)** works on one link only; **global unicast** is internet-wide.

IPv4 → IPv6 transition: dual stack (run both at once), **tunnelling** (carry IPv6 inside IPv4, e.g. 6to4), and **translation** (NAT64). **IPv6 shorthand rules:** drop leading zeros in each group; replace *one* run of all-zero groups with **::**. e.g. 2001:0db8:0000:0000:0000:0000:0000:0001 → 2001:db8::1. **Why IPv6 is faster:** fixed header, no per-router fragmentation, no header checksum.

Memory trick: **IPv4 = 32 bits (4 octets); IPv6 = 128 bits (8 hex groups).** IPv6 has unicast / anycast / multicast — but **no broadcast**. NAT is an IPv4 workaround; IPv6 doesn't need it.

25 Simplex/half/full duplex; sync/async; multiplexing

Q34: Describe the data communication system (with diagram). / Define simplex, half-duplex and full-duplex (examples). / Synchronous vs asynchronous transmission.
✎ **Asked in:** BPSC AP 2022 · BPSC AP 2023 · BKKB AP 2017 · ICT Ministry AP 2017 · 17th NTRCA ICT (2023) · RAKUB ANSE 2023



↑ 5 parts: message, sender, receiver, medium, protocol

Mode	Direction	Example
Simplex	one way only	keyboard, TV broadcast
Half-duplex	both, one at a time	walkie-talkie
Full-duplex	both at once	telephone, switch link

Sync = continuous bit stream with a shared clock (fast, framed); **async** = each byte framed with start/stop bits (simple, slower). **Multiplexing** (FDM/TDM/WDM) sends many signals over one link (BCC AP 2023). **TDMA line** (BTCL AM(Technical) 2021): 8 pairs × 8 kHz × (16+2) bits = 8×8000×18 = **1.152 Mbps**.

Term	meaning	Term	meaning
bit rate	bits per second (bps)	baud rate	signal changes per second
analog signal	continuous wave	digital signal	discrete 0/1 levels
attenuation	signal weakens over distance	noise	unwanted interference

Capacity limits: Nyquist (noiseless) $C = 2 \cdot B \cdot \log_2 L$ bps; **Shannon** (noisy) $C = B \cdot \log_2(1+SNR)$ bps. **FDM** splits by frequency (radio), **TDM** by time slot (digital phone), **WDM** by wavelength (fibre). **Transmission impairments:** attenuation, distortion, noise — fixed by amplifiers/repeaters, shielding and error coding.

Memory trick: **Bit rate ≥ baud rate** (each signal change can carry several bits). Nyquist = noiseless ceiling; Shannon = noisy ceiling. Simplex → Half → Full is "one-way → turn-taking → both-at-once".

26 Parity, checksum & CRC (worked)

Q35: Explain the parity method for error detection and write the ASCII of "Delta". / What is CRC vs parity — which is better?
✎ **Asked in:** BB AME 2023 · RAKUB ANSE 2023 · Combined Bank SO(IT) 2021

Parity adds 1 bit so the total number of 1s is even (even parity) or odd; it catches any single-bit error but misses even numbers of errors. ASCII "Delta" = `68 101 108 116 97` = 01000100 01100101 01101100 01110100

01100001. **CRC beats parity**: it catches burst errors (parity only single-bit). **Simple and two-dimensional parity** both use parity bits (Combined Bank SO(IT) 2021 → answer (d) "a and b").

> **Q36:** CRC: data 11100, divisor 1001 — find the transmitted codeword. / Checksum of (3, 9, 7, 12, 4) in 8-bit — what is actually sent?
> ✎ **Asked in:** Combined Bank AP 2023 · JBL ANA 2022 · Combined Bank AP 2026 · Bank AP written (bank name cut off on the sheet)

```
CRC 11100 / 1001:  append 000 -> 11100000, XOR-divide
   remainder = 111  ->  codeword sent = 11100 111
```

CRC codeword = **11100111** (remainder 111). The receiver divides by 1001; a non-zero remainder = error. Checksum of 3+9+7+12+4 = 35 = 00100011; 1's-complement checksum = **11011100**; sent = the 5 data bytes + this checksum, and the receiver's total should be all-1s.

Checksum adds the data words, takes the 1's complement, and sends it along; the receiver re-adds everything and expects zero (all-ones). Used in IP/TCP/UDP headers (BPDB AE(CSE) 2021, Combined Bank AP 2026).

Method	Detects	Overhead	Where
Simple parity	single-bit (odd count)	1 bit	memory, serial
2-D parity	many single-bit + some bursts	a row + column	blocks
Checksum	upper-layer errors	16 bits	IP/TCP/UDP headers
CRC	burst errors (very strong)	polynomial	Ethernet frame (FCS)

Error correction (vs detection) needs redundancy to fix bits: **Hamming code** uses parity bits at power-of-two positions to locate and flip the wrong bit. **ARQ** (retransmit) is detection + resend; **FEC** (forward error correction) fixes without a resend — used on noisy/one-way links (satellite, deep space).

> **Memory trick:** **Parity** = 1 bit, single-error. **Checksum** = sum + complement, upper layers. **CRC** = polynomial division, catches bursts, used on the wire (Ethernet FCS).

27 Wi-Fi, Bluetooth, WiMAX; CSMA/CD vs CA; 802.x

> **Q37:** Compare Wi-Fi, Bluetooth and WiMAX. / Describe CSMA/CD; why is CSMA/CA used in wireless instead?
> ✎ **Asked in:** NPCBL Engineer(Cyber Security) 2026 · BPSC AME 2021 · BPSC AP 2021 · BOF AE(CSE) 2021 · 47th BCS CSE

	Bluetooth	Wi-Fi	WiMAX
range	**~10 m (PAN)**	~100 m (LAN)	km (MAN)
standard	802.15	802.11	802.16
use	peripherals	home/office LAN	broadband wireless

CSMA/CD (wired Ethernet): listen, send, **detect collisions** and back off. **CSMA/CA** (Wi-Fi): a station *can't* detect collisions on radio (hidden node, half-duplex), so it **avoids** them — sense, wait a random back-off, optionally RTS/CTS, then send (BOF AE(CSE) 2021, BPSC AME 2021). IEEE 802.11 standardises WLAN (Combined Bank SO(IT) 2023). **802.5 = Token Ring** (not Ethernet — Ethernet is 802.3, 49th BCS ICT (2025)).

MCQ as asked	answer	asked in
Shortest-range technology: WiMAX / GSM / Wi-Fi / **Bluetooth**	**Bluetooth**	BTCL JAM 2022
802.16 is the standard for —	**Wireless broadband (WiMAX)**	NWPGCL JAM(ICT) 2022
Max data rate of IEEE 802.11n	**600 Mbps**	Combined Bank SO(IT) 2021
Most-used multiple access in WLAN	**OFDMA**	49th BCS ICT (2025)
Wrong IEEE pairing: **802.5 Ethernet** / 802.11 WLAN / 802.16 WiMAX	**802.5 = Token Ring** (Ethernet = 802.3)	49th BCS ICT (2025)

IEEE 802 family: 802.3 Ethernet · 802.5 Token Ring · 802.11 Wi-Fi · 802.15 Bluetooth/WPAN · 802.16 WiMAX. **Wi-Fi generations:** 802.11n (Wi-Fi 4, 600 Mbps) → 802.11ac (Wi-Fi 5) → 802.11ax (Wi-Fi 6). **4G vs 5G:** 5G has higher bandwidth, far lower latency and uses a wider spectrum than 4G (Combined Bank SO(IT) 2021). **DSSS** (Direct-Sequence Spread Spectrum) spreads a signal over a wide band for noise resistance.

> **Memory trick:** **CD = Collision Detection (wired); CA = Collision Avoidance (wireless).** Radio can't hear itself while sending, so Wi-Fi avoids rather than detects. Range: Bluetooth < Wi-Fi < WiMAX.

28 Unicast/multicast/broadcast; MAC; design & troubleshooting

> **Q38:** Main difference between multicast and broadcast? / Which protocols for connectionless vs connection-oriented? / What is piggybacking and a MAC address?
> ✎ **Asked in:** BPDB AE(CSE) 2021 · BOF AE(CSE) 2021 · Combined Bank Officer(IT) 2026

Cast	To whom	Cast	To whom
Unicast	one host	**Broadcast**	all hosts on the network
Multicast	a group	Anycast	nearest of a group (IPv6)

Multicast → a subscribed group; **broadcast** → everyone (and IPv6 has no broadcast). **Connectionless = UDP; connection-oriented = TCP** (BPDB AE(CSE) 2021). **Piggybacking** = carrying an ACK inside an outgoing data frame to save bandwidth. A **MAC address** = 48-bit hardware address burned into the NIC. Real-time streaming that's connectionless/low-security uses **UDP/RTP** (Combined Bank Officer(IT) 2026).

> **Q39:** Bangladesh Bank WAN links Dhaka & Rangpur offices; low-latency apps and a firewall in front of mail/DNS/web servers — draw the design. / Ping between two directly-connected hosts fails — why?
> ✎ **Asked in:** BB AD(ICT) 2025 · SPCBL SAME 2022

Design: each office LAN → edge router → WAN link (leased line/MPLS/VPN) → the other office; a **firewall** screens the public-facing **DNS/mail/web** servers in a DMZ; low-latency traffic (VoIP) is prioritised (QoS) and uses the shortest path. **Ping fails between two directly-wired hosts** usually because they are in different subnets or need a crossover cable / correct IP+mask on the same subnet. Full-mesh for n offices = n(n−1)/2 WAN links.

> **Exam trap:** Broadcast storms (excess broadcast traffic) come from loops or too-large broadcast domains — segment with **VLANs/routers** and enable **spanning tree** (RAKUB ANSE 2026). A VPN is "a private network over a public network" (BPSC AME 2019).

> **Memory trick:** Casting: **uni** = one, **multi** = a group, **broad** = all, **any** = nearest. A router stops broadcasts; a switch floods them. Piggybacking saves an ACK by riding on a data frame.

★ Night before the exam

1. **IP classes:** A 1–126 (/8), B 128–191 (/16), C 192–223 (/24, 254 hosts), D multicast, E experimental. 127 = loopback.
2. **Private:** 10.0.0.0/8, 172.16.0.0/12, 192.168.0.0/16. **NAT** shares one public IP; not needed in IPv6.
3. **Subnetting:** block = 256 − mask octet; network = multiple of block ≤ IP; broadcast = net + block − 1; usable = block − 2. Subnets = 2^borrowed; hosts = 2^hostbits − 2.
4. **Masks:** /25 .128 (126) · /26 .192 (62) · /27 .224 (30) · /28 .240 (14) · /30 .252 (2). VLSM: biggest block first.
5. **Supernet** = merge contiguous networks (shorter prefix). Routing uses **longest-prefix match**.
6. **OSI (7):** Physical, Data-link, Network, Transport, Session, Presentation, Application. PDU: bit→frame→packet→segment→data.
7. **TCP/IP (4):** Network-access, Internet, Transport, Application. The internet runs TCP/IP, not OSI.
8. **Devices by layer:** hub/repeater L1, switch/bridge L2, router L3, gateway up to L7.
9. **TCP** connection-oriented, reliable, 20-byte header (HTTP/FTP/SMTP); **UDP** connectionless, fast, 8-byte (DNS/VoIP). 3-way handshake: SYN → SYN-ACK → ACK.
10. **Flow control** = protect the receiver; **congestion control** (AIMD) = protect the network.
11. **Routing:** distance-vector (RIP, Bellman-Ford, slow) vs link-state (OSPF, Dijkstra, fast); BGP between ASes.
12. **Topology:** mesh links = n(n−1)/2; star = n; bus = 1. Lab uses star.
13. **Media:** fibre = highest bandwidth, no EMI; UTP twisted to cancel crosstalk; straight-through (unlike) vs crossover (like).
14. **Delay:** transmission = bits ÷ bandwidth; propagation = distance ÷ speed. Nyquist = $2B \cdot \log_2 L$.
15. **IPv4 = 32 bits; IPv6 = 128 bits, 40-byte fixed header, IPsec built-in.** LAN < MAN < WAN.
16. **Errors:** parity (1 bit), checksum (sum + complement), CRC (polynomial division, catches bursts). Hamming corrects.
17. **Wireless:** Bluetooth 802.15 (PAN), Wi-Fi 802.11 (LAN), WiMAX 802.16 (MAN). CD = wired, CA = wireless.
18. **Data-comm:** simplex/half/full duplex; sync vs async; FDM/TDM/WDM multiplexing; Nyquist $2B \cdot \log_2 L$.

19 **Casting:** unicast (one), multicast (group), broadcast (all), anycast (nearest). IPv6 has no broadcast.

✓ Practice MCQs (all asked; answers at the bottom)

1. Which class gives a maximum of 254 hosts? (a) A (b) B (c) C (d) D ✎ BPSC AME 2019

2. The loopback address is: (a) 255.255.255.0 (b) 127.0.0.1 (c) 255.0.0.0 (d) 0.0.0.0 ✎ Combined Bank AHE/AE(IT) 2021

3. How long is an IPv6 address? (a) 32 bits (b) 64 bits (c) 128 bits (d) 256 bits ✎ BPSC AME 2022

4. Maps a known IP to a MAC address: (a) DHCP (b) OSPF (c) ARP (d) NAT ✎ Combined Bank AHE/AE(IT) 2021

5. The OSI transport-layer PDU is a: (a) bit (b) frame (c) packet (d) segment ✎ Combined Bank SO(IT) 2023

6. A router operates at the OSI: (a) physical (b) data-link (c) network (d) transport ✎ 49th BCS CSE (2025)

7. Valid hosts for a /28 network: (a) 16 (b) 8 (c) 15 (d) 14 ✎ BUET IICT April 2021

8. Count-to-infinity occurs in: (a) distance vector (b) SPF (c) link state (d) hierarchical ✎ BPSC AP 2022

9. Which assigns IP, mask, gateway & DNS to a host? (a) NAT (b) DHCP (c) CIDR (d) ISP ✎ Combined Bank SO(IT) 2021

10. Mesh with 5 devices needs how many cables? (a) 5 (b) 6 (c) 10 (d) 15 ✎ BUET M.Sc. April 2021

Answers: 1 c · 2 b · 3 c · 4 c · 5 d · 6 c · 7 d · 8 a · 9 b · 10 c

✎ Written practice (model answer marked)

Q: Given 192.168.2.64/26: find the subnet mask, network, broadcast, first/last usable host and number of valid hosts. (5)

✎ **Asked in:** BUET ICT Cell AHE 2025

/26 → mask 255.255.255.192, block 64 (1).

64 is a multiple of 64 → network 192.168.2.64 (1).

broadcast = 64+64−1 = 192.168.2.127 (1).

first 192.168.2.65, last 192.168.2.126 (1).

valid hosts = 64 − 2 = 62 (1).

Memory trick: Every subnetting answer = **block → network → broadcast → range → usable**. Block = 256 − last mask octet; usable = block − 2.

Before the exam ✓: the magic masks (128/192/224/240/252), OSI order + PDUs, TCP vs UDP, IPv4 vs IPv6 bits, device layers, mesh = n(n−1)/2, transmission vs propagation delay.